Methods for Proportions and Nominal Variables

IV. Contingency Table
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   C. Example
   D. More Than 2x2 Contingency Table
   E. Example
   F. Subdividing Contingency Tables
   G. Comparing One Set of Categorical Observations Against Expected Frequencies: Goodness of Fit χ^2 Test

NO DISCUSSION (Fall Recess)

Summary of Some Statistical Methods to Test Hypotheses

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‡Or interval data that are not necessarily normally distributed.

z Statistic

Table 5-1 Thrombus Formation in People Receiving Dialysis and Treated with Placebo or Aspirin

	Number of patients		
Sample group	Developed thrombi	Free of thrombi	Treated
Placebo	18	7	25
Aspirin	6	13	19
Total	24	20	44

Source: H. R. Harter, J. W. Burch, P. W. Majerus, N. Stanford, J. A. Delmez, C. B. Anderson, and C. A. Weerts "Prevention of Thrombosis in Patients on Hemodialysis by Low-Dose Aspirin," *N. Engl. J. Med.*, **301**:577–579, 1979. Reprinted by permission of the *New England Journal of Medicine.*

$$\hat{p}_{asp} = \frac{6}{19} = 0.316 \qquad \hat{p}_{pla} = \frac{18}{25} = 0.720 \qquad \hat{p} = \frac{6+18}{19+25} = 0.\overline{54}$$

$$z = \frac{\left|\hat{p}_{asp} - \hat{p}_{pla}\right| - \frac{1}{2}\left(1/n_{asp} + 1/n_{pla}\right)}{\sqrt{\hat{p}\left(1-\hat{p}\right)\left(1/n_{asp} + 1/n_{pla}\right)}} = 2.36$$

$$|z| = 2.36 > t_{0.05}(\nu \to \infty) = 1.96 \qquad \text{Reject } H_0 \quad P < 0.02$$

(Using a two-tailed t table) $t_{0.02}(\nu \to \infty) = 2.3263$

IV. Contingency Table

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Under Null Hypothesis of NO Treatment Effect:

A. The Idea

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Under Null Hypothesis of NO Treatment Effect:

24/44 developed thrombi (54.5%) regardless of group
20/44 no thrombi (45.5%) regardless of group

Expected Thrombosis Formation if Aspirin has NO Effect

Sample Group	Expected Develop thrombi	Expected Free of thrombi	Treated
Placebo	(0.545*25)13.64	(0.455*25)11.36	25
Aspirin	(0.545*19)10.36	(0.455*19) 8.64	19
Total	24	20	44

Expected Thrombosis Formation if Aspirin has NO Effect

Sample Group	Expected Develop thrombi	Expected Free of thrombi	Treated
Placebo	(0.545*25)13.64	(0.455*25)11.36	25
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B. Chi-Squared Statistic

$$\chi^2 = \sum\sum \frac{\left(observed - expected\ number\ of\ individuals\ in\ cell\right)^2}{expected\ number\ of\ individuals\ in\ cell}$$

$$\chi^2 = \sum\sum \frac{\left(O-E\right)^2}{E} \qquad \nu = 1$$

$$\chi^2 = \sum\sum \frac{\left(\left|O-E\right| - \frac{1}{2}\right)^2}{E}$$ (Yates's correction)

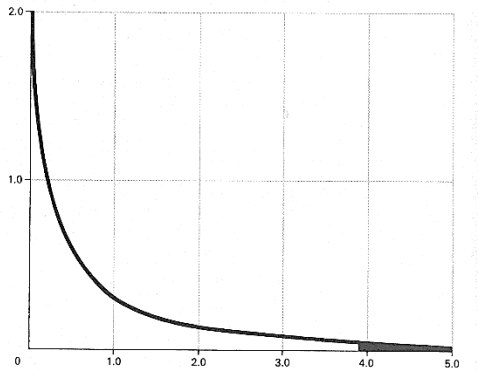


Figure 5-7 The chi-square distribution with 1 degree of freedom. The shaded area denotes the biggest 5 percent of possible values of the χ^2 test statistic

df	$\chi^2_{.995}$	$\chi^2_{.990}$	$\chi^2_{.975}$	$\chi^2_{.950}$	$\chi^2_{.900}$	$\chi^2_{.100}$	$\chi^2_{.050}$	$\chi^2_{.025}$	$\chi^2_{.010}$	$\chi^2_{.005}$
1	0.000	0.000	0.001	0.004	0.016	2.706	3.841	5.024	6.635	7.879
2	0.010	0.020	0.051	0.103	0.211	4.605	5.991	7.378	9.210	10.597
3	0.072	0.115	0.216	0.352	0.584	6.251	7.815	9.348	11.345	12.838
4	0.207	0.297	0.484	0.711	1.064	7.779	9.488	11.143	13.277	14.860
5	0.412	0.554	0.831	1.145	1.610	9.236	11.070	12.833	15.086	16.750
6	0.676	0.872	1.237	1.635	2.204	10.645	12.592	14.449	16.812	18.548
7	0.989	1.239	1.690	2.167	2.833	12.017	14.067	16.013	18.475	20.278
8	1.344	1.646	2.180	2.733	3.490	13.362	15.507	17.535	20.090	21.955
9	1.735	2.088	2.700	3.325	4.168	14.684	16.919	19.023	21.666	23.589
10	2.156	2.558	3.247	3.940	4.865	15.987	18.307	20.483	23.209	25.188
11	2.603	3.053	3.816	4.575	5.578	17.275	19.675	21.920	24.725	26.757
12	3.074	3.571	4.404	5.226	6.304	18.549	21.026	23.337	26.217	28.300
13	3.565	4.107	5.009	5.892	7.042	19.812	22.362	24.736	27.688	29.819
14	4.075	4.660	5.629	6.571	7.790	21.064	23.685	26.119	29.141	31.319
15	4.601	5.229	6.262	7.261	8.547	22.307	24.996	27.488	30.578	32.801

$$\chi^2 > \chi^2_\alpha(\nu) \qquad \text{Reject } H_0 \qquad P < \alpha$$

C. Placebo-Aspirin Example

$$\chi^2 = \sum\sum \frac{\left(|O-E|-\frac{1}{2}\right)^2}{E}$$

(Yates's continuity correction)

$$\chi^2 = \frac{\left(|18-13.64|-\frac{1}{2}\right)^2}{13.64} + \frac{\left(|7-11.36|-\frac{1}{2}\right)^2}{11.36} + \frac{\left(|6-10.36|-\frac{1}{2}\right)^2}{10.36} + \frac{\left(|13-8.64|-\frac{1}{2}\right)^2}{8.64} = 5.57$$

$$\chi^2 = 5.57 > \chi^2_{0.05}(\nu = 1) = 3.841$$

$$\chi^2 = 5.57 > \chi^2_{0.025}(\nu = 1) = 5.024$$

Reject H_0 $P < 0.025$

D. More Than 2x2 Contingency Table

	Outcome				
Treatment	1	2	…	n_c	Total
Group 1	O_{11}	O_{12}	…		R_1
Group 2	O_{21}	O_{22}	…		R_2
…					…
Group n_r	O_{nr1}	…			R_{nr}
	C_1	C_2			n

$$E_{ij} = \frac{C_j}{n} R_i \qquad \chi^2 = \sum_{i=1}^{n_r} \sum_{j=1}^{n_c} \frac{\left(O_{ij} - E_{ij}\right)^2}{E_{ij}} \qquad v = \left(n_r - 1\right)\left(n_c - 1\right)$$

$$\chi^2 > \chi^2_\alpha(v) \qquad \text{Reject } H_0 \qquad P < \alpha$$

(use Yates's correction only if 2x2)

E. Example Nefazadone, Psychotherapy, N&P on Depression

	Outcome		
Treatment	Depression	No Depression	Total
N	36	131	167
P	41	132	173
N&P	75	104	179
	152	367	$519 = n$

H_0: There is no difference in the incidence of depression among the three treatment groups

E. Example Nefazadone, Psychotherapy, N&P on Depression

Outcome

Treatment	Depression	No Depression	Total
N	36	131	167
P	41	132	173
N&P	75	104	179
	152	367	$519 = n$

Expected Results if NO Treatment Effect

Treatment	Depression	No Depression	Total
N	48.9	118.1	167
P	50.7	122.3	173
N&P	52.4	126.6	179
	152	367	$519 = n$

$$E_{ij} = \frac{C_j}{n} R_i$$

$$\chi^2 = \sum_{i=1}^{n_r} \sum_{j=1}^{n_c} \frac{\left(O_{ij} - E_{ij}\right)^2}{E_{ij}} \qquad v = \left(n_r - 1\right)\left(n_c - 1\right)$$

$$\chi^2 = 21.2 \qquad v = \left(3-1\right)\left(2-1\right) = 2$$

Table 5-7 Critical Values for the χ^2 Distribution

	Probability of greater value P							
ν	.50	.25	.10	.05	.025	.01	.005	.001
1	.455	1.323	2.706	3.841	5.024	6.635	7.879	10.828
2	1.386	2.773	4.605	5.991	7.378	9.210	10.597	13.816
3	2.366	4.108	6.251	7.815	9.348	11.345	12.838	16.266
4	3.357	5.385	7.779	9.488	11.143	13.277	14.860	18.467
5	4.351	6.626	9.236	11.070	12.833	15.086	16.750	20.515

$$\chi^2 = 21.2 > \chi^2_{0.001}(\nu = 2) = 13.82$$

reject H_0 with $P < 0.001$

F. Subdividing Contingency Table into 2x2 Sub-Tables

Compare N vs. P

Observed

	Depression	**No Depression**	*total*
N	36	131	*167*
P	41	132	*173*
total	*77*	*263*	***340***

Expected

	Depression	**No Depression**	*total*
N	$\frac{77}{340}\cdot 167 = 37.8$	$\frac{263}{340}\cdot 167 = 129.2$	*167*
P	$\frac{77}{340}\cdot 173 = 39.2$	$\frac{263}{340}\cdot 173 = 133.8$	*173*
total	*77*	*263*	***340***

Subdividing Contingency Table into 2x2 Sub-Tables

Compare N vs. N+P

Observed

	Depression	**No Depression**	*total*
N	36	131	*167*
N+P	75	104	*179*
total	*111*	*235*	***346***

Expected

	Depression	**No Depression**	*total*
N	$\frac{111}{346} \cdot 167 = 53.6$	$\frac{235}{346} \cdot 167 = 113.4$	*167*
N+P	$\frac{111}{346} \cdot 179 = 57.4$	$\frac{235}{346} \cdot 179 = 121.6$	*179*
total	*111*	*235*	***346***

Subdividing Contingency Table into 2x2 Sub-Tables

Compare P vs. N+P

Observed

	Depression	**No Depression**	*total*
P	41	132	*173*
N+P	75	104	*179*
total	*116*	*236*	***352***

Expected

	Depression	**No Depression**	*total*
P	$\frac{116}{352} \cdot 173 = 57.0$	$\frac{236}{352} \cdot 173 = 116.0$	*173*
N+P	$\frac{116}{352} \cdot 179 = 59.0$	$\frac{236}{352} \cdot 179 = 120.0$	*179*
total	*116*	*236*	***352***

Find χ^2 Value for Each 2x2 Comparison
Use Yates's Correction

Compare N vs. P

$$\chi^2_{NvsP} = \frac{\left(|36-37.8|-\frac{1}{2}\right)^2}{37.8} + \frac{\left(|41-39.2|-\frac{1}{2}\right)^2}{39.2} + \frac{\left(|131-129.2|-\frac{1}{2}\right)^2}{129.2} + \frac{\left(|132-133.8|-\frac{1}{2}\right)^2}{133.8} = 0.117$$

Compare N vs. N+P

$$\chi^2_{NvsN+P} = \frac{\left(|36-53.6|-\frac{1}{2}\right)^2}{53.6} + \frac{\left(|75-57.4|-\frac{1}{2}\right)^2}{57.4} + \frac{\left(|131-113.4|-\frac{1}{2}\right)^2}{113.4} + \frac{\left(|104-121.6|-\frac{1}{2}\right)^2}{121.6} = 15.5$$

Compare P vs. N+P

$$\chi^2_{PvsN+P} = \frac{\left(|41-57.0|-\frac{1}{2}\right)^2}{57.0} + \frac{\left(|75-59.0|-\frac{1}{2}\right)^2}{59.0} + \frac{\left(|132-116.0|-\frac{1}{2}\right)^2}{116.0} + \frac{\left(|104-120.0|-\frac{1}{2}\right)^2}{120.0} = 12.4$$

$\nu = 1$ For all three pairwise comparisons because each sub-table is 2x2

Use Bonferroni or Holm Correction to Control Overall Significance of Pairwise Comparisons

Order χ^2 values in descending order and compare to critical value

Calculated χ^2 value	j	α_j (Holm)	$\chi^2_{\alpha_j}(\nu=1)$
$\chi^2_{NvsN+P}=15.5$	1	$\frac{0.01}{(3-1+1)}=0.00\overline{333}$	$\chi^2_{0.00\overline{3}}<\chi^2_{0.001}=10.828$
$\chi^2_{PvsN+P}=12.4$	2	$\frac{0.01}{(3-2+1)}=0.005$	$\chi^2_{0.005}=7.879$
$\chi^2_{NvsP}=0.117$	3	$\frac{0.01}{(3-3+1)}=0.01$	$\chi^2_{0.01}=6.635$

k = number of pairwise comparisons = 3

$\alpha_T = 0.01 =$ *overall significance level*

for Holm correction, $\alpha_j = \frac{\alpha_T}{(k-j+1)}$

Use Bonferroni or Holm Correction to Control Overall Significance of Pairwise Comparisons

$$\chi^2_{NvsN+P} = 15.5 > 10.828 = \chi^2_{0.001} > \chi^2_{0.003}$$

Reject null hypothesis of no difference and move on to next comparison

$$\chi^2_{PvsN+P} = 12.4 > 7.879 = \chi^2_{0.005}$$

Reject null hypothesis of no difference and move on to next comparison

$$\chi^2_{NvsP} = 0.117 \not> 6.635 = \chi^2_{0.01}$$

Do not reject null hypothesis of no difference

Conclusion: response to N vs. N+P is different, and response to P vs. N+P is different, and response to N vs. P is not different with $P < 0.01$

note: $\nu = 1$ for all of these pairwise comparisons

G. Comparing One Set of Categorical Observations Against Expected Frequencies: Goodness of Fit χ^2 Test

- Suppose we have a single set of categorical data and wish to compare the distribution of data points in each category to a set of expected frequencies (a "goodness of fit" test)
  The expected frequencies may come from a prior study/hypothesis
  There are N data points and k categories in this single data set

- H0: There is no difference between the observed and expected distribution of values in each category

OR

- H0: $p_{1,\text{Obs}} = p_{1,\text{Exp}}$, $p_{2,\text{Obs}} = p_{2,\text{Exp}}$, etc.
  (where p_i is the proportion of values in category i)

$$E_i = p_{i,\text{Exp}} \cdot N \qquad \chi^2 = \sum_{i=1}^{k} \frac{(O_i - E_i)^2}{E_i} \qquad \nu = k - 1$$

G. Comparing One Set of Categorical Observations Against Expected Frequencies: Goodness of Fit χ^2 Test

Example:

- A university surveyed graduating seniors about their experiences as undergrads
  - 60% reported getting no regular exercise during their undergraduate career
  - 25% reported getting sporadic exercise
  - 15% reported exercising regularly

- The university wished to increase the amount of exercise its students get while matriculating and launched a health-promotion campaign

- Four years later, the university surveyed the current graduating seniors and obtained the following results:

	No Regular Exercise	**Sporadic Exercise**	**Regular Exercise**	**Total**
Number of Students	255	125	90	470

Based on these data, is there a difference in students' exercise behavior after the campaign?

Example adapted from https://sphweb.bumc.bu.edu/otlt/mph-modules/bs/bs704_hypothesistesting-chisquare/index.html

G. Comparing One Set of Categorical Observations Against Expected Frequencies: Goodness of Fit χ^2 Test

$$E_i = p_{i,\text{Exp}} \cdot N \qquad \chi^2 = \sum_{i=1}^{k} \frac{(O_i - E_i)^2}{E_i} \qquad \nu = k - 1$$

	No Regular Exercise	**Sporadic Exercise**	**Regular Exercise**	**Total**
Observed	255	125	90	470
Expected	$470 \cdot (0.6) = 282$	$470 \cdot (0.25) = 117.5$	$470 \cdot (0.15) = 70.5$	470

$$\chi^2 = \frac{(255 - 282)^2}{282} + \frac{(125 - 117.5)^2}{117.5} + \frac{(90 - 70.5)^2}{70.5} = 8.457$$

$$\nu = 3 - 1 = 2 \qquad \chi^2{}_{0.05}(2) = 5.991 \qquad \chi^2{}_{0.025}(2) = 7.378$$

$\chi^2 = 8.457 > \chi^2{}_{0.025} = 7.378$; reject H0 with $P < 0.025$ and conclude that the exercise behavior of the current graduating seniors is different from 4 years prior

Example adapted from https://sphweb.bumc.bu.edu/otlt/mph-modules/bs/bs704_hypothesistesting-chisquare/index.html

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